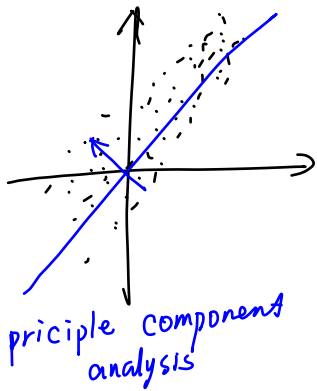


Analytic Geometry 3: Projection of vector.

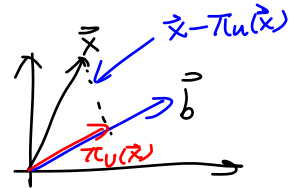
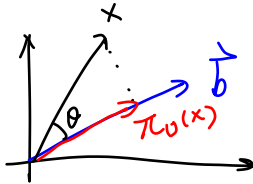


- In machine learning, there are a lot of high dimensional data. $\rightarrow$ hard to analyze
- only a few dimension is of interest.
- Try to project the original high-dimensional data on to a lower dimensional space.
- How to project in a correct way



1. projection: Let V be a vector space and $U \subseteq V$ be a subspace of V .
A linear mapping $\pi: V \rightarrow U$ is called a projection if $\pi^2 = \pi \circ \pi = \pi$
transformation matrix $P_\pi \Rightarrow P_\pi^2 = P_\pi \cdot P_\pi = P_\pi$

Remark: Projecting n times $\Leftrightarrow$ projecting once.



2. Projection onto 1-D space.

(1) $\pi_U(\vec{x})$ is an element U that is closest to $\vec{x}$, i.e.

1) $\exists \lambda \in \mathbb{R}, \pi_U(\vec{x}) = \lambda \vec{b}$, with $\vec{b}$ being the basis vector of U

2) inner product: $\langle \vec{x} - \pi_U(\vec{x}), \vec{b} \rangle = 0$

(2) What is λ ? 1), 2) $\Rightarrow \langle \vec{x} - \lambda \vec{b}, \vec{b} \rangle = 0$

$$\Leftrightarrow \langle \vec{x}, \vec{b} \rangle - \lambda \langle \vec{b}, \vec{b} \rangle = 0 \quad ?$$

bilinear map.

$$f(\lambda \vec{x} + \mu \vec{y}, \vec{z}) = \lambda f(\vec{x}, \vec{z}) + \mu f(\vec{y}, \vec{z})$$

$$\Leftrightarrow \lambda = \frac{\langle \vec{x}, \vec{b} \rangle}{\langle \vec{b}, \vec{b} \rangle} = \frac{\vec{b}^T \vec{x}}{\vec{b}^T \vec{b}} \quad \checkmark$$

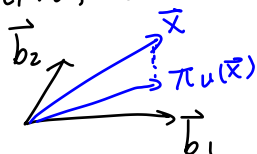
(3) norm of projection $\pi_U(\vec{x})$

$$\|\pi_U(\vec{x})\| = \|\lambda \vec{b}\| = \left\| \frac{\vec{b}^T \vec{x}}{\|\vec{b}\|^2} \vec{b} \right\| = \frac{|\vec{b}^T \vec{x}|}{\|\vec{b}\|^2} \|\vec{b}\| = \frac{\|\vec{x}\| \cdot \|\vec{b}\| \cos \theta}{\|\vec{b}\|^2} \|\vec{b}\| = |\cos \theta| \|\vec{x}\|$$

(4) What is P_π ?

$$\pi_U(\vec{x}) = \lambda \vec{b} = \vec{b} \cdot \lambda = \vec{b} \cdot \frac{\vec{b}^T \vec{x}}{\|\vec{b}\|^2} \Rightarrow P_\pi = \frac{\vec{b} \vec{b}^T}{\|\vec{b}\|^2}$$

3. Projection onto m -dimensional space ($m \geq 1$)



$\Rightarrow$ extension from 1-D case

(1) $\pi_U(\vec{x})$ is an element on U that is closest to $\vec{x}$

$$1) \exists \lambda_i \in \mathbb{R}, i \in [1, m], \pi_U(\vec{x}) = \sum_{i=1}^m \lambda_i \vec{b}_i$$

2) inner product $\vec{x} - \pi_U(\vec{x}) \perp \{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_m\}$, i.e.

$$\begin{cases} \langle \vec{b}_1, \vec{x} - \pi_U(\vec{x}) \rangle = \vec{b}_1^T (\vec{x} - \pi_U(\vec{x})) = 0 \\ \vdots \\ \langle \vec{b}_m, \vec{x} - \pi_U(\vec{x}) \rangle = \vec{b}_m^T (\vec{x} - \pi_U(\vec{x})) = 0 \end{cases}$$

let $B = [\vec{b}_1 \dots \vec{b}_m]$

$$\pi_U(\vec{x}) = \sum_{i=1}^m \lambda_i \vec{b}_i = B \vec{\lambda} \quad \vec{\lambda} = [\lambda_1 \lambda_2 \dots \lambda_m]^T$$

$$\vec{b}_1^T (\vec{x} - B \vec{\lambda}) = 0$$

$$\vec{b}_m^T (\vec{x} - B \vec{\lambda}) = 0$$

$$\Leftrightarrow \begin{bmatrix} \vec{b}_1^T \\ \vdots \\ \vec{b}_m^T \end{bmatrix} [\vec{x} - B \vec{\lambda}] = \vec{0} \Leftrightarrow B^T (\vec{x} - B \vec{\lambda}) = \vec{0}$$

$$\Leftrightarrow B^T \vec{x} = B^T B \vec{\lambda}$$

since B consists of basis of U (linear independent) $\Rightarrow B$ has full rank.

$$B^T B \text{ is invertible} \Rightarrow \vec{\lambda} = (B^T B)^{-1} B^T \vec{x}$$

$$\Rightarrow \pi_U(\vec{x}) = B \vec{\lambda} = B (B^T B)^{-1} B^T \vec{x} \Rightarrow \text{projection matrix } P_U = B (B^T B)^{-1} B^T$$

$$= \frac{B B^T}{B^T B}$$

Remark: $\text{rank}(A^T A) = \text{rank}(A)$ for any matrix $A \in \mathbb{R}^{n \times n}$.

According to Rank-Nullity Theorem, for any $A \in \mathbb{R}^{n \times m}$,
 $\text{rank}(A) + \text{nullity}(A) = n$. $A \vec{x} = \vec{0}$

we want to show $\text{rank}(A) = \text{rank}(A^T A)$ by showing
 $\text{null}(A) = \text{null}(A^T A)$ ✓

How to prove set $A=B$? $\Rightarrow A \subseteq B$ and $B \subseteq A$

① showing $\text{null}(A) \subseteq \text{null}(A^T A)$

for any $\vec{x} \in \text{null}(A)$, we need to show $\vec{x} \in \text{null}(A^T A)$

$$\Rightarrow A \vec{x} = \vec{0} \text{ , one has } A^T (A \vec{x}) = \vec{0} \Rightarrow A^T A \vec{x} = \vec{0} \text{ , } \vec{x} \in \text{null}(A^T A)$$

$$\Rightarrow \text{null}(A) \subseteq \text{null}(A^T A) \text{ (a)}$$

② show $\text{null}(A^T A) \subseteq \text{null}(A)$

for any $\vec{x} \in \text{null}(A^T A)$, $A^T A \vec{x} = \vec{0}$, therefore $\vec{x}^T A^T A \vec{x} = 0 = (A \vec{x})^T A \vec{x} = \|A \vec{x}\|^2 = 0$

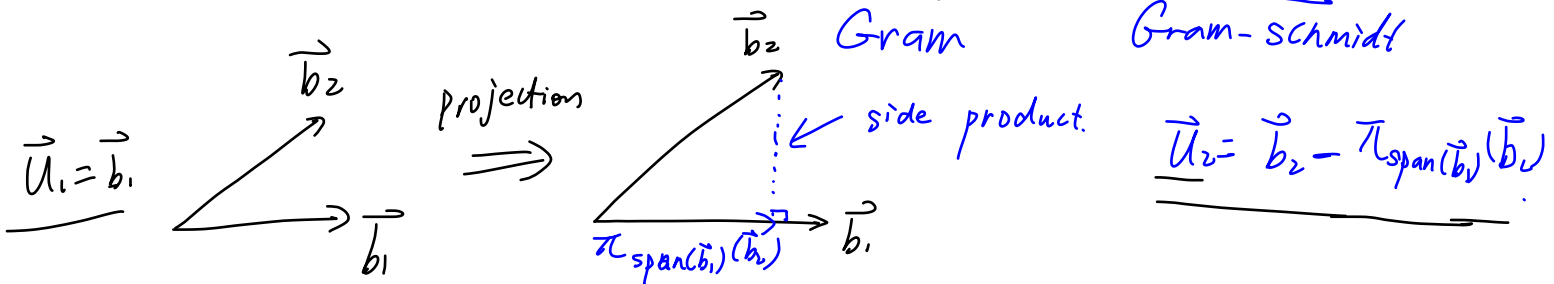
$$\Rightarrow A \vec{x} = \vec{0} \Rightarrow \vec{x} \in \text{null}(A) \Rightarrow \text{null}(A^T A) \subseteq \text{null}(A) \text{ (b)}$$

$$(a) \text{ (b)} \Rightarrow \text{null}(A) = \text{null}(A^T A)$$

consider any B , we need to compute $(B^T B)^{-1}$, which is expensive, unless $B^T B = I$, meaning that B contains an orthonormal basis.

2. Gram - schmidt Orthogonalization

Goal: Transforming a basis $(\vec{b}_1, \dots, \vec{b}_n)$ of an n -dimensional vector space V into an orthogonal / orthonormal basis of V



key idea: find a vector that is perpendicular to $\vec{u}_1$ and $\vec{u}_2$, projection,!

$$\vec{u}_3 = \vec{b}_3 - \pi_{\text{span}(\vec{u}_1, \vec{u}_2)}(\vec{b}_3) \quad \text{Let } M = [\vec{u}_1, \vec{u}_2]$$

$$= \vec{b}_3 - M(M^T M)^{-1} M^T \vec{b}_3 = \vec{b}_3 - \lambda M M^T \vec{b}_3$$

or
$$\vec{u}_3 = \vec{b}_3 - \frac{\vec{u}_1 \vec{u}_1^T}{\|\vec{u}_1\|^2} \vec{b}_3 - \frac{\vec{u}_2 \vec{u}_2^T}{\|\vec{u}_2\|^2} \vec{b}_3 \quad (\text{can be computed recursively})$$

$\Rightarrow$ for n -dimensional basis, we obtain a set of orthogonal basis based on $[\vec{b}_1, \dots, \vec{b}_n]$ by recursively doing

$$\vec{u}_1 = \vec{b}_1$$

$$\vec{u}_2 = \vec{b}_2 - \pi_{\text{span}(\vec{u}_1)}(\vec{b}_2) = \vec{b}_2 - \pi_{\vec{u}_1}(\vec{b}_2)$$

$\vdots$

$$\vec{u}_k = \vec{b}_k - \pi_{\text{span}(\vec{u}_1, \dots, \vec{u}_{k-1})}(\vec{b}_k) = \vec{b}_k - \sum_{i=1}^{k-1} \pi_{\vec{u}_i}(\vec{b}_k)$$

then normalize $\vec{u}_1, \dots, \vec{u}_n$ by

$$\vec{e}_i = \frac{\vec{u}_i}{\|\vec{u}_i\|}$$

Example $\vec{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, compute an orthonormal basis with Gram-Schmidt orthogonalization

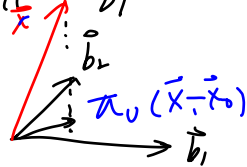
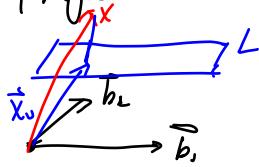
answer: $\vec{u}_1 = \vec{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\vec{u}_2 = \vec{b} - \frac{\vec{a} \vec{a}^T}{\|\vec{a}\|^2} \vec{b} = \vec{b} - \frac{\vec{a}^T \vec{b}}{\|\vec{a}\|^2} \vec{a}$

$$= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{(\sqrt{3})^2} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\vec{u}_1^T \vec{u}_2 = 1 \times 0 + 1 \times (-1) + 1 \times 1 = 0 \quad \vec{u}_1 \perp \vec{u}_2$$

normalizations $\vec{e}_1 = \frac{\vec{u}_1}{\|\vec{u}_1\|} = \frac{[1 \ 1 \ 1]^T}{\sqrt{3}} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \vec{e}_2 = \frac{\vec{u}_2}{\|\vec{u}_2\|} = \frac{[0 \ -1 \ 1]^T}{\sqrt{2}} = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

4. Projection onto an affine space -



$$U = \text{span}(\vec{b}_1, \vec{b}_2)$$

$$L = \vec{x}_0 + U$$

step 1: project $\vec{x} - \vec{x}_0$ onto U .

$$\pi_U(\vec{x} - \vec{x}_0)$$

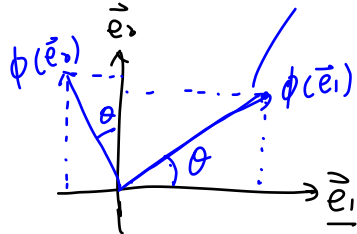
step 2: Translate $\pi_U(\vec{x} - \vec{x}_0)$

back to L by adding

$$\Rightarrow \pi_L(\vec{x}) = \vec{x}_0 + \pi_U(\vec{x} - \vec{x}_0) \rightarrow \vec{x}_0 = \vec{x}$$

5. Rotation:

(1) Rotation in $\mathbb{R}^2$. consider a standard basis $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$



$$\phi(\vec{e}_1) = [\cos\theta, \sin\theta]^T$$

$$\phi(\vec{e}_2) = [-\sin\theta, \cos\theta]^T$$

$$R(\theta) = [\phi(\vec{e}_1) \ \phi(\vec{e}_2)] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Rotation matrix

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad R(\theta) \vec{e}_1 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$\Rightarrow$ Rotation is a linear mapping

In 2-D case: given rotation angle θ , vector $\vec{x}$.

$$\vec{x}_{\text{new}} = R(\theta) \vec{x}$$

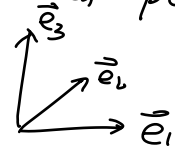
$\uparrow$ positive direction $\rightarrow$ counter-clockwise

(2) Rotation in $\mathbb{R}^3$

we can consider the rotation of any two dimensional plane about 1-D axis

(1) Rotation about $\vec{e}_1$ -axis

$$R_1(\theta) = [\phi(\vec{e}_1) \ \phi(\vec{e}_2) \ \phi(\vec{e}_3)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$



(2) rotating about $\vec{e}_2$ -axis

$$R_2(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

(3) rotating about $\vec{e}_3$ -axis.

$$R_3(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

General rotation can be treated as combination of 3 of the basic rotation above.

3) property of rotations.

(1) preserve distance: $\forall \vec{x}, \vec{y} \in V \quad \|\vec{x} - \vec{y}\| = \|R_0(\vec{x}) - R_0(\vec{y})\|$

(2) preserve angle: angle between $\vec{x}$ and $\vec{y}$ =
angle between $R_0(\vec{x})$ and $R_0(\vec{y})$

(3) for $n \geq 3$, rotation matrices are NOT commutative.

$\forall \vec{x} \in \mathbb{R}^n \quad n \geq 3, \quad R(\theta_1)R(\theta_2)\vec{x} \neq R(\theta_2)R(\theta_1)\vec{x}$, in general

Section 3.8 - 3.9